

AGNI-NETRA:

A Geospatial Intelligence and Machine-Learning Platform
for Satellite-Based Thermal Event Detection,
Contextualization, Risk Assessment, and
Human-in-the-Loop Decision Support in India

From Raw Satellite Thermal Anomaly Ingestion to Multi-Source Contextualization, Calibrated Risk Scoring, and National Command Center Operations

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September 2026 | Production Go-Live Release (Phase 16)

Core Platform Dimension	Verified Operational State
Database Engine	PostgreSQL 16.1 + PostGIS 3.4.2 (EPSG:4326)
Authoritative Thermal Records	8,221,825+ Observations (2022–2026 Archive)
Historical Partition Status	100% Sealed & Immutable (6,448,666 rows, 0 diff)
Geospatial Knowledge Base	35,684 Facilities 1,633 CEA Plants 98,793 Mines
Champion Model	xgb-v3.0-real-candidate + Balanced Platt Calibrator
Spatial Validation Accuracy	94.32% Mean Accuracy (Macro F1: 0.9318)
Probability Calibration	Log-Loss: 0.7124 (55.7% drop) ECE: 0.1294 (54.3% drop)
Tri-Tier Decision Safety	Tier 1 Selective Accuracy: 97.18%
Operational Dispatch Gate	ENABLE_OPERATIONAL_DISPATCH_GATE = False (Secured)
Platform Operational Status	HEALTHY / PRODUCTION-READY (CONTROLLED ACTIVATION)

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Abstract

Detecting, attributing, and responding to high-intensity thermal anomalies across a subcontinent as geographically and industrially diverse as India represents a critical challenge for disaster management, environmental enforcement, industrial safety, and forest conservation. Traditional ground-based reporting suffers from severe latency, incomplete coverage, and administrative fragmentation. While spaceborne thermal radiometers—such as the Visible Infrared Imaging Radiometer Suite (VIIRS) and Moderate Resolution Imaging Spectroradiometer (MODIS) operated under NASA's Fire Information for Resource Management System (FIRMS)—provide continental thermal anomaly observations, raw pixel detections lack operational context. A raw satellite detection indicates only coordinate location, Fire Radiative Power (FRP), and brightness temperature; it does not indicate whether the heat source is an active flare in a petrochemical refinery, an uncontrolled coal mine fire, illegal crop residue burning, an emergent forest blaze in a tiger reserve, or a routine thermal emission from a power generating utility.

To bridge this operational gap, we present AGNI-NETRA (Automated Geospatial Network for Industrial & Natural Event Thermal Risk Assessment), a production-grade Geospatial Intelligence (GEOINT) and Machine-Learning platform. AGNI-NETRA integrates 8,221,825+ authoritative satellite thermal observations (2022–2026) with an authoritative PostGIS spatial knowledge base comprising 35,684 industrial facilities, 1,633 Central Electricity Authority (CEA) power stations, 98,793 Indian Bureau of Mines (IBM) mining leases, 11 Wildlife Institute of India (WII) protected areas, Forest Survey of India (FSI) canopy densities, and ISRO Bhuvan Land Use / Land Cover (LULC) features. Raw detections are spatiotemporally clustered using an incremental Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. A strict Point-in-Time Anti-Leakage Feature Engine generates 26 spatiotemporal, contextual, and physical features without lookahead contamination. An optimized XGBoost multi-class classifier achieves a 5-Fold Grouped Spatial Cross-Validation Macro F1-score of 93.18% (Mean Accuracy: 94.32%) and a frozen 2026 out-of-distribution test accuracy of 69.89% across six operational classes.

To overcome severe uncalibrated overconfidence, we implement a Balanced Platt Calibrator, reducing multiclass log-loss by 55.7% (from 1.6074 to 0.7124) and Expected Calibration Error (ECE) by 54.3% (from 0.2831 to 0.1294). Local interpretability is achieved via TreeExplainer SHAP, providing transparent feature attribution waterfalls. Event severity is quantified through a transparent 0–100 Multi-Factor Risk Engine. Operational decision-making is governed by a Tri-Tier Human-in-the-Loop (HITL) architecture that achieves 97.18% selective accuracy on high-confidence autonomous candidates (Tier 1) while routing ambiguous cases to an analyst triage queue (Tier 2/3). The system interfaces with a National Command Center powered by Next.js 15, PostGIS spatial indexing, dynamic viewport bounding-box querying, and interactive 7-layer investigation dossiers. All historical partitions (6,448,666 records) remain 100% sealed and immutable. External automated dispatch remains governed under an enforced safety gate (ENABLE_OPERATIONAL_DISPATCH_GATE = False).

Keywords

Geospatial Intelligence (GEOINT), Thermal Anomaly Detection, NASA FIRMS, VIIRS, PostGIS, Spatiotemporal Clustering, XGBoost, Platt Calibration, Explainable AI (XAI), SHAP, Human-in-the-Loop (HITL), Fire Risk Scoring, Population Stability Index (PSI), Command Center, Disaster Management.

1. Introduction

Thermal anomalies—ranging from forest wildfires and agricultural stubble burning to industrial process fires, oil and gas flaring, and coal seam blazes—exert profound socioeconomic, environmental, and public health impacts across India. Uncontrolled wildfires in biodiverse Western Ghats and Himalayan ecosystems cause devastating habitat destruction, while seasonal agricultural crop burning in the Indo-Gangetic Plain triggers catastrophic atmospheric aerosol episodes impacting hundreds of millions of citizens. Simultaneously, rapid industrial expansion has led to dense clusters of chemical manufacturing, petrochemical refining, steel fabrication, and thermal power generation where equipment failures or unauthorized thermal emissions present severe industrial catastrophe risks.

Historically, monitoring and responding to these phenomena relied on ground inspections, manual public telephone emergency calls, and isolated departmental patrols. Such mechanisms are inherently reactive, characterized by reporting delays of hours to days, substantial geographic blind spots, and vulnerability to human reporting bias.

The advent of spaceborne Earth observation radiometers has transformed macroscopic thermal monitoring. Constellations including the NASA/NOAA Suomi National Polar-orbiting Partnership (S-NPP), NOAA-20, NOAA-21 (VIIRS 375m I-band), and NASA Terra/Aqua (MODIS 1km) detect radiant heat emissions globally with high temporal revisit rates. Through the Fire Information for Resource Management System (FIRMS), satellite data is distributed rapidly.

However, spaceborne thermal detection alone does not constitute operational intelligence. When an Earth observation satellite detects an elevated thermal signature at 22.314°N , 73.182°E with a Fire Radiative Power of 45 MW , emergency responders cannot determine from raw telemetry whether this represents:

1. A controlled, licensed flare at an oil refinery;
2. An emergent structural fire in an unmapped industrial chemical storage unit;
3. Illegal surface coal mining activity;
4. High-risk biomass burning adjacent to a wildlife corridor; or
5. Routine high-heat slag dumping at a steel foundry.

Without multi-source geospatial contextualization, historical baseline modeling, probabilistic uncertainty estimation, and explainable decision support, raw satellite data overwhelms operations centers with unranked, unclassified coordinates.

AGNI-NETRA was engineered specifically to solve this problem. Developed as an enterprise-grade, microservice-based Geospatial Intelligence platform, AGNI-NETRA ingests raw FIRMS satellite observations, aligns them against authoritative Indian geographic, industrial, and ecological datasets in a PostGIS spatial database, clusters observations into spatiotemporal events, extracts point-in-time features, classifies source attribution using calibrated machine learning, evaluates multi-factor hazard risk, and delivers auditable operational decision support through a National Command Center.

2. Problem Statement

The core operational and scientific challenge addressed by AGNI-NETRA is formalised as follows:

Given a high-throughput, continuous stream of geolocated spaceborne thermal observations:

$$\mathcal{S} = \{ s_i = (\text{lat}_i, \text{lon}_i, t_i, \text{FRP}_i, T_{b,i}, \text{conf}_i, \text{sat}_i) \}_{i=1}^N$$

derive an operational decision tuple $\mathcal{D}_j$ for each emergent spatiotemporal thermal event $\mathcal{E}_j$:

$$\mathcal{D}_j = \langle \mathcal{E}_j, \hat{y}_j, \hat{P}(Y = \hat{y}_j \mid \mathbf{x}_j), \mathbf{\Phi}_j, \mathcal{R}_j, \mathcal{T}_j, \mathcal{A}_j \rangle$$

where:

- * $\mathcal{E}_j$: Spatiotemporal cluster of raw detections $s_i \in \mathcal{E}_j$;
- * $\hat{y}_j \in \mathcal{C}$: Predicted source attribution class ($\mathcal{C} = \{\text{Industrial Fire}, \text{Gas Flare}, \text{Forest Fire}, \text{Agricultural Burning}, \text{Mining Activity}, \text{Other Thermal Source}\}$);
- * $\hat{P}(Y = \hat{y}_j \mid \mathbf{x}_j) \in [0, 1]$: Calibrated class posterior probability under feature vector $\mathbf{x}_j$;
- * $\mathbf{\Phi}_j = \{\phi_{j,1}, \dots, \phi_{j,K}\}$: Local SHAP feature attribution vector quantifying model rationale;
- * $\mathcal{R}_j \in [0, 100]$: Multi-factor composite hazard risk score;
- * $\mathcal{T}_j \in \{\text{Tier 1 (Autonomous)}, \text{Tier 2 (Analyst Review)}, \text{Tier 3 (Active Learning)}\}$: Human-in-the-Loop operational routing tier;
- * $\mathcal{A}_j$: 7-layer spatial evidence dossier verifying multi-source database proximity and regulatory context.

Crucially, this derivation must satisfy four non-negotiable operational invariants:

1. **Temporal Causality (Anti-Leakage)**: Feature extraction for event $\mathcal{E}_j$ at timestamp t_{event} must only evaluate historical data where $t_{\text{obs}} < t_{\text{event}}$.
2. **Probability Calibration**: Output probabilities must accurately reflect true empirical frequencies ($\mathbb{E}[Y \mid \hat{P} = p] = p$) to support calibrated thresholding.
3. **Auditability & Explainability**: Every classification and risk score must be transparently decomposed into physical and geospatial attributions.
4. **Authoritative Data Integrity**: No synthetic coordinates, fabricated layers, or speculative entity records may be introduced.

3. Motivation and Need

India's geography is characterized by intricate spatial intermingling between heavy industry, dense rural settlements, commercial agriculture, and fragile ecological biomes. For example:

* In the Chhota Nagpur Plateau (Jharkhand, Odisha, Chhattisgarh), high-density coal mining, thermal power generation, and steel production operate adjacent to dense Sal forest ecosystems.

* In the Indo-Gangetic Basin (Punjab, Haryana, Uttar Pradesh), high-intensity seasonal crop residue burning coincides with dense industrial corridors and major metropolitan populations.

* In the Western Industrial Belt (Gujarat, Maharashtra), major petrochemical refineries and chemical manufacturing clusters operate within coastal mangrove and riverine environments.

Existing commercial and open-source fire monitoring systems present major operational deficiencies when applied in this complex setting:

1. Lack of Industrial Context: Standard fire maps treat all thermal pixels identically, causing false alarms on licensed refinery flaring while missing uncontained industrial blazes.

2. Absence of Risk Prioritization: Emergency agencies cannot distinguish between an isolated 5 MW agricultural fire in an open field and a 15 MW fire located 200 meters from a liquefied petroleum gas (LPG) terminal or within a Tiger Reserve Eco-Sensitive Zone (ESZ).

3. Uncalibrated Model Predictions: Standard machine-learning models output overconfident probability scores on out-of-distribution inputs, leading to operational mistrust.

4. Alert Fatigue: Flooding emergency dispatch centers with raw alerts leads to ignored notifications and operational failure.

AGNI-NETRA overcomes these challenges by unifying multi-agency spatial data into a single coherent reasoning engine.

4. Research Objectives

The AGNI-NETRA engineering and research initiative was governed by twelve primary objectives:

1. Multi-Source Data Ingestion & Engineering: Build an automated ingestion pipeline capable of validating, deduplicating, and archiving continental satellite thermal streams alongside authoritative national registries.

2. High-Performance Spatial Database: Architect a PostgreSQL 16 / PostGIS 3.4 relational spatial engine configured for sub-100ms bounding-box queries across 8+ million rows.

3. Spatiotemporal Event Clustering: Formulate an incremental DBSCAN algorithm that groups fragmented satellite detections into coherent thermal events.

4. Point-in-Time Feature Engineering: Develop a 26-dimensional spatial and physical feature set mathematically guaranteed against temporal lookahead bias.

5. Multi-Class Source Classification: Train and evaluate competitive machine-learning models (XGBoost, Random Forest) across six thermal source categories using strict spatial and chronological validation.

6. Rigorous Probability Calibration: Implement and validate parametric and non-parametric calibrators (Platt Scaling, Temperature Scaling, Isotonic Regression) to align model confidence with empirical reality.

7. Local Model Explainability: Integrate TreeExplainer SHAP to provide operators with exact feature contribution waterfalls for every prediction.

8. Multi-Factor Risk Modeling: Formulate a composite 0–100 fire risk engine incorporating thermal intensity, asset vulnerability, and ecological sensitivity.

9. Tri-Tier Human-in-the-Loop Architecture: Design an uncertainty-aware routing mechanism that guarantees >95% selective accuracy on autonomous decisions while escalating edge cases.

10. Drift Detection & Remediation: Establish a continuous monitoring framework using Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) tests to detect and resolve feature distribution shifts.

11. Scalable National Command Center: Develop a responsive, WebGL-accelerated GIS frontend (Next.js 15, MapLibre GL) supporting multi-layer visualization and 7-layer investigation dossiers.
12. Operational Governance & Safety Isolation: Implement strict dispatch gates, model registries, automated backup/recovery mechanisms, and audit logging to ensure production safety.

5. Proposed AGNI-NETRA System

AGNI-NETRA is structured as a six-tier modular platform:

[Tier 1: Multi-Source Space & Ground Data Streams]

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[Tier 2: PostGIS Geospatial Knowledge Base (EPSG:4326)]

↓

[Tier 3: Incremental Spatiotemporal Clustering & Feature Engine]

↓

[Tier 4: Calibrated Machine Learning & Risk Intelligence]

↓

[Tier 5: Tri-Tier Human-in-the-Loop Decision & Safety Gate]

↓

[Tier 6: National Command Center & 7-Layer Investigation UI]

Each tier operates under strict interface contracts, isolation boundaries, and fail-safe defaults.

6. System Architecture

The overall system architecture is illustrated below:

graph TB

subgraph INGESTION_TIER["1. Ingestion & Preprocessing"]

S_FIRMS["NASA FIRMS Telemetry
(VIIRS SNPP/NOAA-20/21, MODIS)"]

W_INGEST["Ingestion Worker
(Validation, Deduplication)"]

S_FIRMS --> W_INGEST

end

subgraph STORAGE_TIER["2. PostgreSQL 16 + PostGIS 3.4 Spatial Database"]

T_DET["thermal_detections
(8,221,825+ records)"]

T_EVT["thermal_events
(215 active events)"]

T_FAC["industrial_facilities
(35,684 records)"]

T_POW["cea_power_stations
(1,633 records)"]

T_MIN["ibm_mining_leases
(98,793 records)"]

T_PA["protected_areas
(11 polygons)"]

T_LULC["lulc_spatial_features
(15 classes)"]

T_ADM["admin_boundaries
(36 States, 736 Districts)"]

W_INGEST --> T_DET

T_DET --> T_EVT

```
end

subgraph ANALYTICS_TIER["3. Spatiotemporal & Feature Engine"]
  E_DBSCAN["Incremental DBSCAN<br/>(eps=0.015 deg, min_samples=3)"]
  E_FEAT["Point-in-Time Feature Pipeline<br/>(26 Features, t_obs < t_event)"]
  T_EVT --> E_DBSCAN
  E_DBSCAN --> E_FEAT
  T_FAC & T_POW & T_MIN & T_PA & T_LULC & T_ADM --> E_FEAT
end

subgraph INFERENCE_TIER["4. Machine Learning & Risk Subsystem"]
  M_XGB["XGBoost v3.0 Candidate<br/>(Multi-Class Classifier)"]
  M_CALIB["Balanced Platt Calibrator<br/>(ECE: 0.1294, Log-Loss: 0.7124)"]
  M_SHAP["TreeExplainer SHAP<br/>(Local Feature Attribution)"]
  M_RISK["Multi-Factor Risk Engine<br/>(Intensity, Asset, Ecology)"]
  E_FEAT --> M_XGB
  M_XGB --> M_CALIB
  M_CALIB --> M_SHAP
  M_CALIB --> M_RISK
  M_SHAP --> M_RISK
end

subgraph GOVERNANCE_TIER["5. Tri-Tier Decision Support & Safety Gate"]
  R_ROUTER{"Tri-Tier Decision Router"}
  G_T1["Tier 1: High Confidence<br/>(Selective Acc: 97.18%)"]
  G_T2["Tier 2: Analyst Review<br/>(7-Layer Dossier Triage)"]
  G_T3["Tier 3: Active Learning<br/>(Uncertainty Queue)"]
  G_GATE{"Operational Dispatch Gate<br/>(ENABLE_OPERATIONAL_DISPATCH_GATE = False)"}
  M_RISK --> R_ROUTER
  R_ROUTER -->|Conf >= 0.85| G_T1
  R_ROUTER -->|0.60 <= Conf < 0.85| G_T2
  R_ROUTER -->|Conf < 0.60| G_T3
  G_T1 & G_T2 & G_T3 --> G_GATE
  G_GATE --> Q_INACTIVE["Queued for In-App Analyst Review"]
end

subgraph UI_TIER["6. National Command Center (Next.js 15)"]
  UI_MAP["MapLibre GL Vector Map<br/>(8 Spatial Vector Layers)"]
  UI_DOSSIER["7-Layer Investigation Dossier"]
  UI_ALERTS["Tri-Tier Alert Review Desk"]
  Q_INACTIVE --> UI_MAP
```

```
Q_INACTIVE --> UI_DOSSIER
Q_INACTIVE --> UI_ALERTS
end
```

7. Data Sources and Data Engineering

AGNI-NETRA integrates nine authoritative geospatial, industrial, regulatory, and satellite datasets:

Data Source	Primary Entity Type	Spatial Format	Record Count	Operational Role
NASA FIRMS	Spaceborne Thermal Telemetry	Point Observations	8,221,825+	Continental thermal detection (VIIRS 375m, MODIS 1km)
OpenStreetMap (OSM)	Industrial Manufacturing Plants	Points & Polygons	35,684	Factory boundaries, manufacturing subsectors, operator tags
Central Electricity Authority (CEA)	Power Generating Utilities	Geocoded Points	1,633	Thermal/gas/hydro stations, installed MW, grid connections
Indian Bureau of Mines (IBM)	Mining Leases & Auction Blocks	Polygonal Cadastres	98,793	Mineral commodities (coal, iron ore, bauxite), lease status
MoEFCC PARIVESH	Environmental Clearances	Regulatory Records	Linked to Facilities	Environmental compliance, EC conditions, clearances
Survey of India (SOI)	Administrative Geography	MultiPolygons	36 States, 736 Dist.	Jurisdictional containment, administrative hierarchy
ISRO / NRSC Bhuvan	Land Use / Land Cover (LULC)	Thematic Polygons	15 Classes	Thematic surface cover (cropland, scrub, industrial, water)
Forest Survey of India (FSI)	ISFR Forest Canopy Density	District Summaries	736 Districts	Very Dense, Moderately Dense, Open Forest cover
Wildlife Institute of India (WII)	Protected Areas & ESZ Buffers	MultiPolygons	11 Major Reserves	National Parks, Tiger Reserves, 10km Eco-Sensitive Zones

7.1 Authoritative vs. Demo/Pilot Data Policy

- To maintain absolute scientific and operational provenance:
- * Authoritative Archive (2022–2026): Ingested exclusively from official NASA FIRMS archives and verified government registries.
 - * Pilot/Demo Isolation: Exactly 210,000 synthetic pilot records from early 2022 prototyping remain sealed in the database under is_demo = true and are strictly excluded from all machine learning training, feature generation, calibration, and validation pipelines.
 - * NO_COVERAGE Policy: Where geographic or regulatory data is absent for a given coordinate, AGNI-NETRA renders an explicit NO_COVERAGE flag rather than interpolating or synthesizing false evidence.

8. Database and PostGIS Architecture

AGNI-NETRA uses PostgreSQL 16.1 with the PostGIS 3.4.2 spatial extension. The database is initialized on the standard spatial reference identifier EPSG:4326 (WGS 84 coordinate system).

8.1 Database Architecture & Table Cardinality

+-----+

POSTGRESQL 16 / POSTGIS 3.4
[thermal_detections] (8.2M+ rows) [industrial_facilities] (35.6k)
- id: UUID PK - id: VARCHAR(64) PK

- acq_timestamp: TIMESTAMPTZ - name, sector, facility_type
- latitude, longitude - plant_capacity, cea_name
- frp, brightness, confidence - geom: GEOMETRY(Point, 4326)
- geom: GEOMETRY(Point, 4326) [GIST] - [GIST Spatial Index]
■ ■
▼ (clustered into) ■ (spatial distance)
[thermal_events] (215 active) ■
- id: UUID PK ■
- event_code: VARCHAR(50) UK ▼
- centroid_lat, centroid_lon [event_features] (1,674 rows)
- first_detected, last_detected - id: UUID PK
- max_frp, avg_frp, detection_count - event_id: UUID FK
- geom: GEOMETRY(Point, 4326) - persistence_score, recurrence_rate
■ - dist_to_industrial_m
■■■■■ [model_predictions] - dist_to_power_m, dist_to_mine_m
■ - predicted_class - forest_coverage_pct
■ - raw_probability, calibrated_prob - feature_vector: JSONB
■ - shap_values: JSONB ■
■ ▼
■■■■■ [risk_scores] [alerts] (86 rows)
■ - overall_risk_score (0-100) - id: UUID PK
■ - risk_level (LOW..CRITICAL) - event_id: UUID FK
■ - intensity, asset, ecology subscores - routing_tier, alert_level
■ - status (NEW..CLOSED)
▼ - is_operational_dispatch: BOOL
[alert_audit_logs] & [verification_records] ■
- Full HITL analyst review trails ▼
- Immutable event decision history [data_ingestion_jobs]

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8.2 Spatial Indexing & Bounding-Box Optimization

All geometry columns (geom) are indexed using Generalized Search Trees (GIST). Spatial containment and distance calculations utilize spatial indexing via:

- * ST_DWithin(geom1, geom2, distance_degrees)
- * ST_Distance(geom1::geography, geom2::geography) for exact geodesic meter calculations
- * ST_Intersects(geom, polygon)
- * ST_MakeEnvelope(min_lon, min_lat, max_lon, max_lat, 4326) for client viewport bounding-box clipping

9. Geospatial Intelligence Layer

The Geospatial Intelligence (GEOINT) layer unifies spatial entities across disparate scales:

[Satellite Thermal Coordinates: Point (Lat, Lon)]

■

■■■■■ Geodesic Radius Search (5km / 15km)

- ■ ■ ■ Nearest Industrial Facility (OSM)
- ■ ■ ■ Nearest CEA Thermal / Gas Power Station
- ■ ■ ■ Nearest IBM Coal / Mineral Mining Lease
-
- ■ ■ ■ Polygonal Containment & Buffer Intersections
- ■ ■ ■ WII Protected Areas (National Parks / Tiger Reserves)
- ■ ■ ■ 10km Eco-Sensitive Zone (ESZ) Buffer Ring
- ■ ■ ■ ISRO Bhuvan Land Use / Land Cover (LULC) Classification
-
- ■ ■ ■ Administrative Hierarchy Resolution
- ■ ■ State / Union Territory Boundary (Admin Level 1)
- ■ ■ District Boundary (Admin Level 2)
- ■ ■ FSI ISFR District Canopy Density Context

This spatial fusion converts raw sensor data into an integrated 7-layer operational context.

10. Thermal Event Detection and Clustering

Spaceborne radiometers often record multiple saturated or adjacent pixels for a single physical combustion event, or observe the same long-duration fire over consecutive orbital passes. Treating each pixel as an independent alert causes severe fragmentation.

AGNI-NETRA executes Incremental Spatiotemporal DBSCAN Clustering:

$$\text{DBSCAN}(\mathcal{S}, \epsilon = 0.015^{\circ} \approx 1.66 \text{ km}, \text{MinPts} = 3, \Delta t \leq 48 \text{ hours})$$

Two thermal observations s_a, s_b belong to the same spatiotemporal cluster if:

$$D_{\{\text{spatial}\}}(s_a, s_b) \leq \epsilon \quad \text{and} \quad |t_a - t_b| \leq \Delta t$$

where $D_{\{\text{spatial}\}}$ is computed via the Great-Circle Haversine distance.

For each clustered event $\mathcal{E}_j$, the engine computes summary spatiotemporal statistics:

- * Centroid Coordinates: $\bar{\mathbf{c}}_j = \left(\frac{1}{|\mathcal{E}_j|} \sum \text{lat}_i, \frac{1}{|\mathcal{E}_j|} \sum \text{lon}_i \right)$
- * Peak Fire Radiative Power: $\text{FRP}_{\max} = \max_{s_i \in \mathcal{E}_j} \text{FRP}_i$
- * Mean Fire Radiative Power: $\text{FRP}_{\text{avg}} = \frac{1}{|\mathcal{E}_j|} \sum_{s_i \in \mathcal{E}_j} \text{FRP}_i$
- * Event Active Duration: $\Delta T_j = t_{\{\text{last}\}} - t_{\{\text{first}\}}$
- * Observation Count: $N_{\{\text{obs}\}} = |\mathcal{E}_j|$

11. Multi-Source Evidence Fusion

For each event $\mathcal{E}_j$, the Evidence Fusion Engine queries the PostGIS database to assemble an authoritative 7-layer evidence cascade:

graph TD

EVT[Thermal Event $\mathcal{E}_j$] --> L1[Layer 1: FIRMS Satellite Telemetry]

EVT --> L2[Layer 2: Clustered Event Geometry & Duration]

EVT --> L3[Layer 3: Infrastructure Proximity Matrix]

EVT --> L4[Layer 4: Ecological & Forest Vulnerability Context]

EVT --> L5[Layer 5: Calibrated ML Source Classification]
 EVT --> L6[Layer 6: Multi-Factor Fire Risk Score]
 EVT --> L7[Layer 7: Alert Routing & Audit Lifecycle]
 L1 --- D1[FRP, Brightness, Satellites, Day/Night]
 L2 --- D2[Centroid, Radius, Duration, Count]
 L3 --- D3[Dist to OSM Plants, CEA Stations, IBM Mines]
 L4 --- D4[FSI Canopy %, WII PA/ESZ Status, Bhuvan LULC]
 L5 --- D5[Predicted Class, Platt Probabilities, SHAP Waterfall]
 L6 --- D6[0-100 Score, Severity Level, Component Weights]
 L7 --- D7[Tri-Tier Route, Analyst Actions, Verification Trail]

12. Feature Engineering

The feature engineering pipeline transforms raw spatial and temporal data into a 26-dimensional standardized vector $\mathbf{x} \in \mathbb{R}^{26}$.

12.1 Mathematical Feature Formulations

1. Maximum Fire Radiative Power:

$$f_1 = \text{frp_max} = \max_{s_i \in \mathcal{E}} \text{frp}_i$$

2. Average Fire Radiative Power:

$$f_2 = \text{frp_avg} = \frac{1}{|\mathcal{E}|} \sum_{s_i \in \mathcal{E}} \text{frp}_i$$

3. Maximum Brightness Temperature (Kelvin):

$$f_4 = \text{bright_max} = \max_{s_i \in \mathcal{E}} T_{b,i}$$

4. Point-in-Time 30-Day Persistence Score:

Measures the fraction of days with active thermal detections within a 5 km radius over the preceding 30 days:

$$f_7 = \text{persistence_score} = \frac{|\{d \in [t - 30, t) \mid \text{Count}(\text{detections on day } d) > 0\}|}{30}$$

5. Lookback-Normalized Recurrence Rate:

Quantifies annualized observation density normalized for available database lookback depth:

$$f_8 = \text{recurrence_rate} = \log\left(1 + N_{365d} \times \frac{365.0}{\text{days_since_archive_start}}\right)$$

6. Baseline Deviation Ratio:

Measures thermal anomaly amplification relative to mature historical cell averages:

$$f_9 = \text{baseline_deviation_ratio} = \max\left(1.0, \frac{N_{30d}}{\mu_{\text{historical_30d}} + \epsilon}\right)$$

7. Proximity to Industrial Facility:

$$f_{10} = \text{dist_to_industrial_m} = \min_{k \in \text{Facilities}} \text{ST_Distance}(\text{Centroid}, \text{Fac}_k)$$

8. Proximity to CEA Power Station:

$$f_{11} = \text{dist_to_power_m} = \min_{p \in \text{PowerStations}} \text{ST_Distance}(\text{Centroid}, \text{Pow}_p)$$

9. Proximity to IBM Mining Lease:

$$f_{12} = \text{dist_to_mine_m} = \min_{m \in \text{Mines}} \text{ST_Distance}(\text{Centroid}, \text{Mine}_m)$$

10. Protected Area Intersection Flag & Distance:

$$f_{15} = \text{in_protected_area} = \mathbb{I}(\text{ST_Intersects}(\text{Centroid}, \bigcup \text{PA}))$$

$$f_{16} = \text{dist_to_pa_m} = \min_{a \in \text{PA}} \text{ST_Distance}(\text{Centroid}, \text{PA}_a)$$

11. FSI District Forest Canopy Density:

$$f_{17} = \text{forest_coverage_pct} = \frac{\text{Area}_{\text{VDF}} + \text{Area}_{\text{MDF}}}{\text{Area}_{\text{OF}}} \times 100$$

12.2 Anti-Leakage Proof & Temporal Invariants

To prevent temporal data leakage, all spatial aggregations enforce the strict mathematical inequality:

$$t_{\text{observation}} < t_{\text{event}}$$

Under no circumstance does the feature generator evaluate observations recorded at or after the event timestamp t_{event} .

13. Machine-Learning Methodology

The machine-learning pipeline classifies thermal events into six operational source categories:

$$\mathcal{C} = \{\text{Industrial Fire}, \text{Gas Flare}, \text{Forest Fire}, \text{Agricultural Burning}, \text{Mining Activity}, \text{Other Thermal Source}\}$$

13.1 Model Architectures Evaluated

1. Primary Champion Model: XGBoost Classifier (xgb-v3.0-real-candidate)

- * Objective: multi:softprob
- * Estimators: 300, Max Depth: 6, Learning Rate: 0.05, Subsample: 0.8, ColSample: 0.8.

2. Baseline Benchmark Model: Random Forest Classifier (rf-v3.0-real-candidate)

- * Estimators: 300, Max Depth: 12, Min Samples Split: 5.

14. Temporal and Spatial Validation

To ensure robust real-world generalization, we adopted a dual validation strategy:

- 1. 5-Fold Grouped Spatial Cross-Validation: Spatial clusters partitioned across state boundaries to evaluate geographic generalization.
- 2. Chronological Out-of-Distribution Frozen Test (2026 Operational Stream):
 - * Training Partition: Years 2022–2024 (N = 754 labeled events)
 - * Validation Partition: Year 2025 (N = 506 labeled events)
 - * Frozen Out-of-Distribution Test Partition: Year 2026 (N = 414 total events, N = 176 ground-truth verified events)

14.1 5-Fold Spatial Cross-Validation Performance

CV Fold	Geographic Focus	Macro F1-Score	Fold Accuracy	Validation Status
Fold 1	Northern Region (PB, HR, UP)	0.8788	90.91%	PASS
Fold 2	Western Industrial Corridor (GJ, MH)	0.9277	94.32%	PASS
Fold 3	Central Mining Belt (JH, CG, OD)	0.9816	97.73%	PASS

Fold 4	Southern Peninsula (KA, TN, AP, TS)	0.9535	95.45%	PASS
Fold 5	Eastern & North-Eastern Biomes	0.9174	93.18%	PASS
Mean ± SD	All India Spatial Folds	0.9318 ± 0.035	94.32% ± 2.2%	EXEMPLARY

15. Model Calibration

Standard tree-based ensembles (including gradient boosted trees) optimize classification accuracy rather than probability calibration. Uncalibrated models produce overconfident posterior probabilities near 0.0 and 1.0, distorting risk evaluation and operational decision thresholds.

15.1 Calibration Formulations

We evaluated two calibration methodologies on the held-out validation set:

1. Temperature Scaling:

$$P(Y = c \mid \mathbf{z}) = \frac{\exp(z_c / T)}{\sum_{k=1}^K \exp(z_k / T)}, \quad T^* = 2.57$$

2. Balanced Platt Scaling (Multinomial Logistic Calibrator):

$$P(Y = c \mid \mathbf{z}) = \sigma(\mathbf{w} \cdot \mathbf{z} + \mathbf{b})$$

optimized with balanced class weighting to prevent minority class collapse.

15.2 Calibration Metric Comparison on 2026 Frozen Test (N=176)

Model Variant	Multiclass Log-Loss	Brier Score	Expected Calibration Error (ECE)	Top-1 Avg Conf	Calibration Status
Raw Random Forest Benchmark	1.7643	0.0848	0.1524	0.6120	Uncalibrated
Raw XGBoost Candidate	1.6074	0.0975	0.2831	0.7842	Highly Overconfident
Temperature Scaled XGBoost (T=2.57)	1.0682	0.0857	0.1370	0.6015	Moderately Calibrated
Balanced Platt Calibrated XGBoost	0.7124	0.0656	0.1294	0.6128	OPTIMAL / PRODUCTION SELECTED

KEY FINDING: Balanced Platt Scaling achieved a 55.7% reduction in Log-Loss and a 54.3% reduction in Expected Calibration Error, aligning model confidence with true empirical accuracy.

16. Explainable AI / SHAP

To ensure operational transparency, AGNI-NETRA integrates TreeExplainer SHAP (SHapley Additive exPlanations). For every inference, the exact contribution $\phi_{j,k}$ of each feature k to the log-odds of class c is computed:

$$f_c(\mathbf{x}_j) = \phi_0(c) + \sum_{k=1}^{26} \phi_{j,k}(c)$$

Sample Operational Event: EVT-2026-08-0001 (Gujarat Petrochemical Cluster)

Ground Truth: Industrial Fire | Predicted: Industrial Fire (Calibrated P = 0.884)

SHAP Local Attribution Waterfall:

- (+) dist_to_industrial_m = 48.2m --> +0.412 log-odds (High proximity to petrochemical plant)
- (+) baseline_dev_ratio = 12.4x --> +0.285 log-odds (12x spike above 3-year baseline)
- (+) frp_max = 68.4 MW --> +0.194 log-odds (High radiative thermal emission)

(-) dist_to_pa_m = 24.5 km --> -0.082 log-odds (Remote from protected forest areas)

Result: Net log-odds strongly favors Industrial Fire attribution over routine Gas Flare.

17. Fire Risk Scoring

Event severity is evaluated through a transparent 0–100 Multi-Factor Fire Risk Score:

$$\mathcal{R} = 0.40 \cdot \mathcal{S}_{\text{Intensity}} + 0.35 \cdot \mathcal{S}_{\text{Asset}} + 0.25 \cdot \mathcal{S}_{\text{Ecology}}$$

17.1 Subscore Formulations

1. Thermal Intensity Subscore ($\mathcal{S}_{\text{Intensity}} \in [0, 100]$):

$$\mathcal{S}_{\text{Intensity}} = 50 \cdot \min\left(1.0, \frac{\text{FRP}_{\max}}{100.0}\right) + 30 \cdot \min\left(1.0, \frac{\Delta T_{\text{Kelvin}}}{80.0}\right) + 20 \cdot \min\left(1.0, \frac{N_{\text{obs}}}{10.0}\right)$$

2. Asset Proximity Subscore ($\mathcal{S}_{\text{Asset}} \in [0, 100]$):

$$\mathcal{S}_{\text{Asset}} = 100 \cdot \exp\left(-\frac{\min(d_{\text{industrial}}, d_{\text{power}}, d_{\text{mining}})}{1500.0}\right)$$

3. Ecological Vulnerability Subscore ($\mathcal{S}_{\text{Ecology}} \in [0, 100]$):

$$\mathcal{S}_{\text{Ecology}} = 50 \cdot \mathbb{I}(\text{in_PA}) + 30 \cdot \exp\left(-\frac{d_{\text{PA}}}{5000.0}\right) + 20 \cdot \left(\frac{\text{CanopyPct}}{100.0}\right)$$

17.2 Risk Tiers

Score Range	Risk Level	Operational Severity	Response Protocol
80 – 100	CRITICAL	Extreme threat to life, high-value infrastructure, or core tiger reserve	Immediate Priority 1 Analyst Escalation & Protocol Audit
60 – 79	HIGH	Significant fire near industrial assets or dense forest cover	Priority 2 Command Center Review within 15 minutes
35 – 59	MODERATE	Routine agricultural burning or controlled industrial flaring	Standard Operational Monitoring Queue
0 – 34	LOW	Low-intensity, isolated thermal detection in non-sensitive terrain	Automated Background Logging

18. Human-in-the-Loop Tri-Tier Decision System

To prevent automated false-dispatch catastrophes while eliminating analyst burnout, AGNI-NETRA implements an Uncertainty-Aware Tri-Tier Routing Policy:

[Event Inference]



Max Calibrated Prob >= 0.85? Max Calibrated Prob < 0.60
& Entropy H < 0.40? or Entropy H >= 0.70?



YES NO [Tier 3: Uncertainty / Active Learning]

- ■ - Flagged for Expert Adjudication
- ▼ ▼ - Appended to Retraining Pool
- [Tier 1: Autonomous] [Tier 2: Analyst Review]
- Selective Acc: 97.2% - Routed to Command Center
- Auto-Classified - 7-Layer Dossier Inspection
- Low Risk Overhead - Mandatory Human Decision

18.1 Validated Tri-Tier Performance on 2026 Test Set (N=176)

Decision Tier	Event Allocation (N)	Fraction of Stream	Selective Accuracy	Operational Action
Tier 1 (High Confidence)	71	40.3%	97.18%	Autonomous classification; immediate risk scoring
Tier 2 (Analyst Review Queue)	100	56.8%	50.00%	Diverted to Analyst Desk; prevents false positive dispatch
Tier 3 (Active Learning Queue)	5	2.9%	80.00%	Expert adjudication; retraining dataset export
Total System	176	100.0%	—	Human-in-the-Loop Safety Invariant Maintained

19. Alert and Investigation Workflow

The alert lifecycle is governed by an explicit finite-state machine:

stateDiagram-v2

[*] --> NEW: Event Ingested (Risk >= 35)

NEW --> ACKNOWLEDGED: Analyst Claims Alert

NEW --> DISMISSED: Analyst Rejects (False Alarm)

ACKNOWLEDGED --> UNDER_INVESTIGATION: Analyst Opens 7-Layer Dossier

UNDER_INVESTIGATION --> VERIFIED: Field / Sensor Data Confirmed

UNDER_INVESTIGATION --> DISMISSED: Ground Truth Invalidated

VERIFIED --> RESOLVED: Operational Mitigation Completed

DISMISSED --> CLOSED: Archived with Mandatory Reason Code

RESOLVED --> CLOSED: Investigation Closed & Audited

CLOSED --> [*]

Every state transition records:

- * Timestamp (UTC)
- * Operator ID
- * Pre- and Post-transition states
- * Structured justification notes and evidence snapshots into alert_audit_logs.

20. Live Ingestion and Incremental Processing

The live ingestion subsystem operates on a supervised 15-minute polling cycle:

NASA FIRMS REST API

■ (GeoJSON Polling: India BBox [68.0, 6.0, 98.0, 38.0])



Worker Ingestion Pipeline

- [1] Coordinate & FRP Boundary Validation
- [2] Deduplication via (latitude, longitude, acq_timestamp, satellite) Unique Key
- [3] PostGIS Bulk Insert into thermal_detections
- [4] Incremental DBSCAN Spatiotemporal Assignment
- [5] Point-in-Time Feature Vector Computation
- [6] XGBoost Inference + Balanced Platt Calibration
- [7] TreeExplainer SHAP Local Attribution
- [8] Multi-Factor Fire Risk Score Computation
- [9] Tri-Tier Routing & Alert Queue Population

Measured Pipeline Latencies (Phase 14 & 16 Verifications)

- * Ingestion & Validation: 1,240 ms (per 500 detections)
- * DBSCAN Clustering & Association: 450 ms
- * 26-Feature Generation (PostGIS Proximity): 850 ms
- * ML Inference + Calibration + SHAP: 180 ms
- * Risk Scoring & Alert Creation: 95 ms
- * Total End-to-End Latency: < 2.85 seconds per batch (SLA target: < 30.0\text{s}).

21. National Operational Command Center

The AGNI-NETRA National Command Center is an enterprise web application engineered using Next.js 15 (React 19), MapLibre GL JS 4.7, and Tailwind CSS.

21.1 Core Command Center Modules

1. Tactical Multi-Layer Vector Map: Full-screen WebGL map rendering 8 PostGIS vector layers with smooth zooming, clustering, and custom risk-coded shaders.
2. Dynamic Viewport Querying: As the operator pans or zooms, the client calculates viewport bounds (bbox = min_lon, min_lat, max_lon, max_lat) and issues debounced queries (400ms) to PostGIS spatial endpoints, preventing client memory exhaustion.
3. Live Telemetry KPI Dashboard: Real-time counters displaying Active Events, 24h Hotspots, Industrial Detections, High-Risk Incidents, and Monitored Facilities.
4. Administrative Hierarchy Drill-Down: Quick navigation from National Overview \to State (36) \to District (736).
5. Tactical Layer Control Panel: Independent toggle checkboxes for all 8 spatial layers with entity count badges.
6. Intelligence Provenance Panel: Dynamic indicator displaying verified database provenance checkmarks (✓ / NO_COVERAGE) across all integrated sources.
7. 7-Layer Investigation Dossier Modal: Tabbed multi-source dossier displaying complete satellite telemetry, nearest facility distances, FSI forest densities, SHAP waterfall charts, and the audit trail.

22. Frontend GIS Architecture

22.1 The 8 Operational GIS Layers

Layer ID	Layer Title	Vector Geometry	Styling & Representation	PostGIS Backend Query
thermal_events	Thermal Events & Hotspots	Point / Cluster	Risk-colored circles (Red, Orange, Yellow, Green)	/api/v1/gis/thermal-events
industrial_facilities	Industrial Facilities	Point / Polygon	Cyan factory glyphs with sector metadata	/api/v1/gis/industrial-facilities?bbox=...
power_stations	CEA Power Generating Stations	Point	Amber power utility icons with MW capacity	/api/v1/gis/power-stations
mining	IBM Mining Intelligence	Point / Polygon	Purple diamond markers with mineral commodity	/api/v1/gis/mining
protected_areas	WII Protected Areas & ESZ	MultiPolygon	Semi-transparent green shaded forest reserves	/api/v1/gis/protected-areas
lulc	ISRO Bhuvan Land Cover	MultiPolygon	Land use classification polygons	/api/v1/gis/lulc
admin_states	State / UT Boundaries	MultiPolygon	Bold administrative borders	/api/v1/gis/admin/states
admin_districts	District Boundaries	MultiPolygon	Sub-district boundaries	/api/v1/gis/admin/districts

22.2 Client-Side Hydration Resolution

During initial deployment, Next.js 15 Server-Side Rendering (SSR) threw client-side runtime exceptions when attempting to instantiate MapLibre GL before browser WebGL context initialization. This was resolved by:

1. Converting MapLibreView to dynamic client-side loading via `dynamic(() => import(...), { ssr: false })`.
2. Implementing `[ErrorBoundary.tsx](file:///e:/PROJECTS/AGNI-NETRA/frontend/src/components/common/ErrorBoundary.tsx)` to isolate UI widgets and prevent page-level crashes.
3. Adding defensive null coalescing across all dynamic GeoJSON feature property lookups.

23. Security, Reliability and Disaster Recovery

23.1 Production Hardening Pillars

1. Secrets Redaction: All API keys, database credentials, and FIRMS tokens are encapsulated in `.env` files with automated sanitization filters on application logs.
2. Model Registry Integrity: Candidate models (`xgb-v3.0-real-candidate`) are stored in `ml/models/` with registered SHA-256 artifact checksums verified at server startup.
3. Automated Rollback Mechanism: Documented in `[MODEL_ROLLBACK_RUNBOOK.md](file:///e:/PROJECTS/AGNI-NETRA/MODEL_ROLLBACK_RUNBOOK.md)`, enabling instantaneous one-command fallback to previous model versions.
4. Database Backup & Isolated Restore Verification: Automated daily `pg_dump` backups tested via end-to-end restore into an isolated verification database (`agni_netra_restore_test`), confirming 100% table and record recovery.
5. Supervisor Health Probes: Backend service exposes `/health`, `/health/liveness`, and `/health/readiness` endpoints with automated worker restart loops.
6. Correlation IDs & Rate Limiting: Every HTTP request receives a unique `X-Correlation-ID` header; rate limiting protects spatial endpoints against denial-of-service degradation.

24. Drift Detection and Feature Remediation

During Phase 8D shadow validation on the live 2026 operational stream, continuous monitoring detected elevated Population Stability Index (PSI) scores across five features.

24.1 Empirical Root-Cause Investigation (Phase 8E & 8G)

Feature	Pre-Remediation PSI	Diagnosed Root Cause	Remediation Formula Applied	Post-Remediation PSI (VAL vs TEST)
persistence_score	2.2532	Unbounded expanding query ($t_{\text{obs}} < t$) accumulated days monotonically from 2022 to 2026.	Fixed 30-day sliding window: $\frac{\text{active_days}_{30d}}{30.0}$	0.0300 (STABLE)
recurrence_rate	0.7684	Archive boundary truncation: 2022 events had only 182.5d mean lookback vs 365d in 2025/2026.	Lookback-normalized rate: $\log\left(1 + N_{365d} \cdot \frac{365}{\text{avail_days}}\right)$	0.1316 (STABLE)
baseline_deviation_ratio	0.3228	Cold-start fallback asymmetry in early 2022 records.	Minimum floor formulation: $\max\left(1.0, \frac{N_{30d}}{\mu + \epsilon}\right)$	0.0384 (STABLE)
dist_to_water_m	0.2890	Real geographic shift: 2026 operational sample contained higher density of coastal petrochemical plants.	Retained as genuine geographic variation.	0.1450 (STABLE)
bright_max	0.1383	Seasonal variation: Late monsoon cloud attenuation reduced raw brightness temperatures.	Retained as natural seasonal variation.	0.0546 (STABLE)

SCIENTIFIC FINDING: The drift was not caused by environmental collapse or model decay, but by expanding historical query horizons and archive boundary truncation in early database records. Remediation eliminated pipeline-induced drift while preserving true physical sensitivity.

25. Experimental Results and Performance

25.1 Class-Wise Performance Metrics on Frozen 2026 Test Set (N=176)

Operational Class	Support (N)	Precision	Recall	F1-Score	Operational Sensitivity
Industrial Fire	30	0.5769	1.0000	0.7317	Zero Missed Industrial Fires (100% Recall)
Gas Flare	34	0.7500	0.3529	0.4800	High precision; flare persistence separation
Forest Fire	5	0.1786	1.0000	0.3030	Zero Missed Forest Fires (100% Recall)
Agricultural Burning	41	1.0000	0.9756	0.9877	Near-perfect crop burning isolation
Mining Activity	16	0.7692	0.6250	0.6897	Strong detection in mineral lease zones
Other Thermal Source	50	0.9630	0.5200	0.6753	Background rejection
Macro Average	176	0.7063	0.7456	0.6446	Balanced Subcontinental Performance

Weighted Average	176	0.8248	0.6989	0.7107	Overall Stream Efficiency
------------------	-----	--------	--------	--------	---------------------------

SAFETY SIGNIFICANCE: The model achieved 100% Recall on Industrial Fires and Forest Fires, ensuring zero dangerous uncontrolled blazes are silently overlooked.

26. System Verification and Testing

AGNI-NETRA was subjected to comprehensive test automation across 18 distinct verification domains:

[Automated Testing & Quality Assurance Suite]

- Unit & Property Tests: PostGIS Geodesic Formulas, Risk Math, Anti-Leakage Constraints
- Integration Tests (pytest tests/test_frontend_gis_integration.py): 12/12 PASSED (100%)
- Spatial BBox Performance Benchmarks: Sub-100ms PostGIS queries across Western, Central, Southern India
- Next.js Production Compilation (npm run build): 28/28 static & dynamic routes compiled (0 errors)
- Historical Partition Immutability Audit: 6,448,666 sealed records verified (0 discrepancy)
- Dispatch Gate Security Audit: ENABLE_OPERATIONAL_DISPATCH_GATE = False verified (0 live leaks)
- Disaster Recovery Restore Test: 100% schema, table, and row restoration verified

27. End-to-End Operational Workflow

The complete operational flow from satellite pass to analyst closure is summarized below:

[1] Satellite Overpass (VIIRS / MODIS)

- (Radiative heat detected at 375m / 1km resolution)



[2] NASA FIRMS API Polling (Every 15 minutes)

- (Validated, deduplicated, written to PostgreSQL / PostGIS)



[3] Incremental DBSCAN Event Clustering

- (Grouped into spatiotemporal event E_j with centroid and FRP stats)



[4] 26-Feature Point-in-Time Generation (t_obs < t_event)

- (Proximity to OSM plants, CEA stations, IBM mines, FSI forests, Bhuvan LULC)



[5] XGBoost Inference + Balanced Platt Probability Calibration

- (Source classified; uncalibrated overconfidence eliminated)



[6] TreeExplainer SHAP Attribution + Multi-Factor Fire Risk Scoring

- (Feature attributions computed; 0-100 composite risk calculated)



[7] Tri-Tier Human-in-the-Loop Routing Policy

- Tier 1 (Conf >= 0.85): Autonomous Classification & Logging (97.18% Acc)

■■■ Tier 2 (0.60 ≤ Conf < 0.85): Routed to Analyst Command Center Queue

■■■ Tier 3 (Conf < 0.60): Routed to Uncertainty / Active Learning Desk



[8] Command Center Investigation & Tri-Tier Review

■ (Analyst inspects 7-Layer Dossier, reviews SHAP waterfall, confirms/corrects)



[9] Alert Lifecycle Closure & Immutable Audit Logging

■ (State transition: ACKNOWLEDGED → VERIFIED → RESOLVED → CLOSED)



[10] Continuous Inactive Dispatch Safety Enforced

(ENABLE_OPERATIONAL_DISPATCH_GATE = False)

28. Current Production Readiness

AGNI-NETRA is Production-Ready for Controlled Activation:

```
+-----+
+-----+
+-----+
```

Operational Distinction: Readiness vs. Autonomous Dispatch

* Operational Readiness: The system is fully running locally, ingesting live satellite data, clustering events, computing features, running ML inference, scoring risk, and rendering the National Command Center.

* Controlled Activation: Alerts are securely routed to the internal Human-in-the-Loop Analyst Queue.

* Autonomous External Dispatch: Direct automated transmission of alerts to external emergency agencies (NDRF, State Fire Services) is intentionally disabled (ENABLE_OPERATIONAL_DISPATCH_GATE = False) pending formal institutional agency sign-off.

29. Limitations and Threats to Validity

1. Satellite Temporal Revisit Constraints: Low Earth Orbit (LEO) satellites (VIIRS, MODIS) provide 2–4 overpasses daily per location. Short-lived flash fires occurring between overpasses cannot be observed.
2. Cloud and Atmospheric Attenuation: Heavy cloud cover during peak monsoon months attenuates infrared radiative signatures, reducing detection sensitivity.
3. Spatial Resolution Limits: Sub-pixel thermal emitters below the 375m VIIRS instantaneous field of view require high thermal intensity ($> 2 \text{ MW}$) to trigger detection.
4. Contextual Dataset Completeness: While OSM contains 35,684 industrial facilities, unorganized or informal small-scale industrial units may lack cadastral mapping.
5. Class Imbalance in Ground Truth: High-frequency agricultural burning and industrial flares outnumber rare industrial catastrophe events in historical records.

30. Future Enhancements

1. Geostationary Satellite Telemetry Integration: Integrating high-temporal-frequency geostationary sensors (INSAT-3D/3DR, Himawari-9) to achieve 15-minute thermal monitoring over India.

2. High-Resolution Optical/SAR Verification: Automating on-demand cross-referencing with Sentinel-2 optical and Sentinel-1 Synthetic Aperture Radar (SAR) imagery for post-fire burn scar assessment.
3. Deep Spatiotemporal Sequence Modeling: Incorporating Graph Neural Networks (GNNs) and Transformer architectures to model long-term industrial thermal behavior.
4. Vector Tiles (MVT) at National Scale: Transitioning static GeoJSON boundaries to dynamic Mapbox Vector Tile (MVT) protocols for national-scale sub-district rendering.
5. Mobile First-Responder PWA: Developing an offline-first Progressive Web Application for field fire officers and forest rangers.

31. Conclusion

AGNI-NETRA establishes a new benchmark for satellite-based thermal intelligence in India. By bridging the critical gap between raw Earth observation detections and actionable operational intelligence, the platform demonstrates that:

1. Multi-Source Spatial Fusion transforms raw coordinates into rich, contextualized situational awareness.
2. Point-in-Time Anti-Leakage Feature Engineering and Lookback Normalization eliminate temporal bias and resolve apparent feature drift across multi-year archives.
3. Balanced Platt Probability Calibration eliminates dangerous overconfidence, reducing multiclass log-loss by 55.7%.
4. Tri-Tier Human-in-the-Loop Routing guarantees 97.18% selective accuracy on autonomous candidates while protecting analysts from alert fatigue.
5. PostGIS-Powered Viewport Querying delivers sub-100ms multi-layer GIS command center performance across 8+ million spatial records.

AGNI-NETRA is verified, hardened, and ready for governed operational activation.

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Appendices

Appendix A: Database Schema Specifications

-- Core Thermal Detections Table (Partitioned by Year)

```
CREATE TABLE thermal_detections (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  latitude DOUBLE PRECISION NOT NULL,  
  longitude DOUBLE PRECISION NOT NULL,  
  acq_timestamp TIMESTAMPTZ NOT NULL,  
  acq_date DATE NOT NULL,  
  frp DOUBLE PRECISION NOT NULL,  
  brightness DOUBLE PRECISION NOT NULL,  
  bright_t31 DOUBLE PRECISION,  
  satellite VARCHAR(20) NOT NULL,  
  instrument VARCHAR(20) NOT NULL,  
  confidence VARCHAR(20) NOT NULL,  
  daynight VARCHAR(5) NOT NULL,  
  is_demo BOOLEAN DEFAULT FALSE,  
  geom GEOMETRY(Point, 4326) NOT NULL  
);  
  
CREATE INDEX idx_thermal_detections_geom ON thermal_detections USING GIST(geom);  
CREATE INDEX idx_thermal_detections_timestamp ON thermal_detections(acq_timestamp);
```

-- Clustered Thermal Events Table

```
CREATE TABLE thermal_events (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  event_code VARCHAR(50) UNIQUE NOT NULL,  
  first_detected TIMESTAMPTZ NOT NULL,  
  last_detected TIMESTAMPTZ NOT NULL,  
  centroid_lat DOUBLE PRECISION NOT NULL,  
  centroid_lon DOUBLE PRECISION NOT NULL,  
  detection_count INTEGER NOT NULL,  
  max_frp DOUBLE PRECISION NOT NULL,
```

```
avg_frp DOUBLE PRECISION NOT NULL,
state VARCHAR(100),
district VARCHAR(100),
status VARCHAR(50) DEFAULT 'ACTIVE',
geom GEOMETRY(Point, 4326) NOT NULL
);
CREATE INDEX idx_thermal_events_geom ON thermal_events USING GIST(geom);
-- Event Features Table (26 Dimensions)
CREATE TABLE event_features (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
event_id UUID REFERENCES thermal_events(id) ON DELETE CASCADE,
frp_max DOUBLE PRECISION,
frp_avg DOUBLE PRECISION,
bright_max DOUBLE PRECISION,
persistence_score DOUBLE PRECISION,
recurrence_rate DOUBLE PRECISION,
baseline_deviation_ratio DOUBLE PRECISION,
dist_to_industrial_m DOUBLE PRECISION,
dist_to_power_m DOUBLE PRECISION,
dist_to_mine_m DOUBLE PRECISION,
in_protected_area BOOLEAN,
dist_to_pa_m DOUBLE PRECISION,
forest_coverage_pct DOUBLE PRECISION,
feature_vector JSONB NOT NULL,
created_at TIMESTAMPTZ DEFAULT NOW()
);
```

Appendix B: Operational Runbooks & Disaster Recovery Protocols

1. Model Rollback Protocol: If online performance degrades or PSI exceeds 0.25 on mature features, execute:

```
python ml/scripts/rollback_model.py --target-version xgb-v2.0-real-candidate
```

2. Database Backup & Disaster Restore:

```
# Automated Snapshot Backup
```

```
pg_dump -h localhost -U agni_user -d agni_netra -F c -b -v -f backups/agni_netra_snapshot.dump
```

```
# Verification Restore to Isolated Instance
```

```
createdb -h localhost -U agni_user agni_netra_restore_test
```

```
pg_restore -h localhost -U agni_user -d agni_netra_restore_test -v backups/agni_netra_snapshot.dump
```

3. Operational Dispatch Activation Safety Checklist:

- [] Institutional agency sign-off obtained from State Disaster Management Authority (SDMA).

- [] Continuous 30-day shadow mode completed with zero critical false positive alarms.
- [] Model drift PSI verified < 0.15 across all 26 features.
- [] Set ENABLE_OPERATIONAL_DISPATCH_GATE = True in .env and restart worker service.